

U.G.C. NET Ex

COMPUTER

(Solved with Expl

1. How many strings of 5 digits have the property that the sum of their digits is 7?

(a) 66 (b) 330
(c) 495 (d) 99

Ans. (b) : Partition of 7:

7, 0, 0, 0, 0	6, 1, 0, 0, 0	5, 2, 0, 0, 0
5, 1, 1, 0, 0	4, 3, 0, 0, 0	4, 2, 1, 0, 0
4, 1, 1, 1, 0	3, 3, 1, 0, 0	3, 2, 2, 0, 0
3, 2, 1, 1, 0	3, 1, 1, 1, 1	2, 2, 2, 1, 0
2, 2, 1, 1, 1		

Arrangement of above partition = $5! \mid 4!+5! \mid 3!+5! \mid 3!+5! \mid 2!2!+5! \mid 3!+5! \mid 2!+5! \mid 3!+5! \mid 2!2! + 5! \mid 2!2! + 5! \mid 2!+5! \mid 4!+5! \mid 3!+5! \mid 3!2!$
 $= 5 + 20 + 20 + 30 + 20 + 60 + 20 + 30 + 30 + 60 + 50 + 20 + 10$
 $= 330$

2. Consider an experiment of tossing two fair dice, one black and one red. What is the probability that the number on the black die divides the number on red die?

(a) $\frac{22}{36}$ (b) $\frac{12}{36}$
(c) $\frac{14}{36}$ (d) $\frac{6}{36}$

Ans. (c) : (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6) (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)

Total number of possible pout comes = 14

Probability that the number of black dice divides the number dice = $14/36$

3. In how many ways can 15 indistinguishable fish be placed into 5 different ponds, so that each pond contains at least one fish?

(a) 1001 (b) 3876
(c) 775 (d) 200

Ans. (a) : Number of ways it can be done = number of positive integer solution of

$$\Rightarrow x_1 + x_2 + x_3 + x_4 + x_5 = 15$$

$$\Rightarrow (15-1)C_4 = 14 C_4 = 1001$$

4. Consider the following statements :

S₁: Depth - first search is used to traverse a rooted tree.

S₂: Pre - order, Post-order and In order are used to list the vertices of an ordered rooted tree.

S₃: Huffman's algorithm is used to find an optimal binary tree with given weights.

S₄: Topological sorting provides a labeling such that the parents have larger labels than their children.

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PHYSICS
Explanation Paper-II)

Which of the above statements are **true**?

- (a) (S_1) and (S_2)
- (b) (c) and (S_4)
- (c) (S_1), (S_2) and (S_3)
- (d) (S_1), (S_2), (S_3) and (S_4)

Ans. (d) : All the statements are true, so answer is (d)

5. Consider a Hamiltonian Graph (G) with no loops and parallel edges. Which of the following is true with respect to this Graph (G)?

1. $\deg(v) \geq n/2$ for each vertex of G
 2. $|E(G)| \geq 1/2 (n-1)(n-2) + 2$ edges
 3. $\deg(v) + \deg(w) \geq n$ for every v and w not connected by an edge
- (a) 1 and 2
 - (b) 2 and 3
 - (c) 1 and 3
 - (d) 1, 2 and 3

Ans. (d) : S_1 : Dirac theorem: A simple graph with n vertices ($n \geq 3$) is hamiltonian if every vertex has degree $n/2$ or greater.

S_2 : Maximum number of edge is non-hamilton graph is

$$|E(G)| \leq \binom{n-1}{2} + 1$$

Therefore, minimum number of edges in Hamilton graph is

$$|E(G)| \geq \binom{n-1}{2} + 2$$

$$|E(G)| \geq \frac{(n-1)(n-2)}{2} + 2$$

S_3 : Ore's theorem for hamiltonian graph states that $\deg(v) + \deg(w) \geq n$ for every v and w not connected by an edge.

6. Consider the following statements :

- S_1 : Boolean expressions and logic networks correspond to labeled acyclic digraphs.
- S_2 : Optimal Boolean expressions may not correspond to simplest networks.
- S_3 : Choosing essential blocks first in a Carnage map and then greedily choosing the largest remaining blocks to cover may not give an optimal expression.

Which of these statements(s) is/are **correct**?

- (a) S_1 only
- (b) S_2 only
- (c) S_1 and S_2
- (d) S_1 , S_2 and S_3

Ans. (d) : All are correct.

7. Consider a full - adder with the following input values :

(a) $x = 1, y = 0$ and C_1 (carry input) = 0

(b) $x = 0, y = 1$ and $C_1 = 1$

Compute the values of S (sum) and C_0 (carry output) for the above input values.

(1) $S = 1, C_0 = 0$ and $S = 0, C_0 = 1$

(2) $S = 0, C_0 = 0$ and $S = 1, C_0 = 1$

(3) $S = 1, C_0 = 1$ and $S = 0, C_0 = 1$

(4) $S = 0, C_0 = 1$ and $S = 1, C_0 = 0$

Ans. (a) : A full adder can be implemented in many different ways such as with a custom transistor level circuit or composed of other gates.

One implemented is with $S = A \oplus B \oplus C_{in}$ and

$$C_{out} = (A \cdot B) + (C_{in} (A \oplus B)).$$

Given,

$S_1: x = A = 1, y = B = 0$ and $C_{in} = 0$

$$\text{Then, } S = 1 \oplus 0 \oplus 0 = 1 \oplus 0 = 1$$

$$C = (1 \cdot 0) + (0(1 \oplus 0)) = 0$$

$S_2: x = A = 0, y = B = 1$ and $C_{in} = 1$

$$\text{Then, } S = 0 \oplus 1 \oplus 1 = 0 \oplus 0 = 0$$

$$C = (0 \cdot 1) + (1(0 \oplus 1)) = 0 + 1 = 1$$

8. "If my computations are correct and I pay the electric bill, then I will run out of money. If I don't pay the electric bill, the power will be turned off. Therefore, if I don't run out of money and the power is still on, then my computations are incorrect."

Convert this argument into logical notations using the variables c, b, r, p for propositions of computations, electric bills, out of money and the power respectively. (Where $\neg$ means NOT)

(a) if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(\neg r \wedge p) \rightarrow \neg c$

(b) if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(r \wedge p) \rightarrow c$

(c) if $(c \wedge b) \rightarrow r$ and $\neg p \rightarrow \neg b$, then $(\neg r \wedge p) \rightarrow \neg c$

(d) if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$, then $(\neg r \wedge p) \rightarrow \neg c$

Ans. (a) : $S_1: (c \wedge b) \rightarrow r$

$S_2: (\neg b \rightarrow \neg p)$

Conclusion = $(\neg r \wedge p) \rightarrow \neg c$

same as if $(c \wedge b) \rightarrow r$ and $\neg b \rightarrow \neg p$,

then $(\neg r \wedge p) \rightarrow \neg c$

9. Match the following :

List-I

List-II

a. $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

i. Contra positive

b. $[(p \wedge q) \rightarrow r] \Leftrightarrow [p \rightarrow (q \rightarrow r)]$

ii. Exportation law

- c. $(p \rightarrow q) \Leftrightarrow [(p \wedge \neg q) \rightarrow 0]$ iii. Reduction as absurdum
- d. $(p \leftrightarrow q) \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$ iv. Equivalence

Codes :

	a	b	c	d
(a)	(i)	(ii)	(iii)	(iv)
(b)	(ii)	(iii)	(i)	(iv)
(c)	(iii)	(ii)	(iv)	(i)
(d)	(iv)	(ii)	(iii)	(i)

Ans. (a) :

(a) The contra positive of a conditional statement is formed by negating both the hypothesis & the conclusion, and then interchanging the resulting negations.

ex. $(P \rightarrow Q) \Leftrightarrow (\neg Q \rightarrow \neg P)$

(b) Exportation is a valid rule of replacement in propositional logic

$((P \wedge Q) \rightarrow R) \Leftrightarrow (P \rightarrow (Q \rightarrow R))$

(c) Reduction and absurdum $\Rightarrow$

$(P \rightarrow Q) \Leftrightarrow [(P \wedge Q) \rightarrow 0]$

(d) Logical equality is an operation on two logical values, typically the values of two proposition that produces a value of true & false

$(P \rightarrow Q) \wedge (Q \rightarrow P)$

10. Consider a proposition given as :

" $x \geq 6$, if $x^2 \geq 25$ and its proof as :

If $x \geq 6$, then $x^2 = x \cdot x \geq 6 \cdot 6 = 36 \geq 25$

Which of the following is correct w.r.to the given proposition and its proof?

- (1) The proof shows the converse of what is to be proved
 - (2) The proof starts by assuming what is to be shown
 - (3) The proof is correct and there is nothing wrong
- (a) (1) only (b) (3) only
- (c) (1) and (2) (d) (2) only

Ans. (c) : Let $P : x \geq 6$ $Q : x^2 \geq 25$ then given statement can be written as $Q \rightarrow P$ & its given proof may be written as $P \rightarrow Q$ which is converse of $Q \rightarrow P$

80 (1) is correct
we have to prove P & we are assuming it in our proof so (2) is also true.

So answer is (c)

11. What is the output of the following program?

(Assume that the appropriate directive are included and there is no syntax error)

main ()

{ char S[] = "ABCDEFGH",

print f ("%C", *(&S[3]));

print f ("%s", S + 4);

print f ("%u", S);

/* Base address of S is 1000*/ }

- (a) ABCDEFGH1000
- (b) CDEFGH1000
- (c) DDEFGHH1000
- (d) DEFGH1000

Ans. (d) : S is pointer of character array:

1000	1001	1002	1003	1004	1005	1006	1007	1008
------	------	------	------	------	------	------	------	------

A	B	C	D	E	F	G	H	\0
S[3] is character at index 3 of array S which is 'D' and & (S[3]) is address of 'D' but "%c" printed character at address of 'D' which is 'D' itself. (S + 4) escapes 4 character of array of S then remaining printed till null which string is "EFGH". "%u" printed address of array S which is given as 1000. Therefore final printed is "DEFGH1000".								

12. Which of the following in C++, is inherited in a derived class from base class?

- (a) constructor
- (b) destructor
- (c) data members
- (d) virtual methods

Ans. (c) : Data members in C++ is inherited in a derived class from base class.

Constructor is a member function of a class which initialize objects of a class. In C++, constructor is automatically called when object create. It is special member function of the class.

Destructor is a member function which destructs or deletes an object.

Virtual methods is a method which is redefined (overridden) in derived class.

13. Given that $x = 7.5$, $j = -1.0$, $n = 1.0$, $m = 2.0$ the value of $--x + j == x > n >= m$ is :

- (a) 0
- (b) 1
- (c) 2
- (d) 3

Ans. (a) : $--x + j == x > n >= m$

Here we need to check LHS == RHS. So we get answer as 0 or 1 only. By applying values

$--7.5 + (-1.0) == (7.5 > 1.0) = 2.0$

$(6.5 - 1.0) == (0 > 2.0)$

$(5.5 == 0)$

Return false (0) since 5.5 is not equal to 0.

14. Which of the following is incorrect in C++?

- (a) When we write overloaded function we must code the function for each usage
- (b) When we write function template we code the function only once
- (c) It is difficult to debug macros
- (d) Templates are more efficient than macros

Ans. (d) : First 3 choice are correct, & so it 4th Templates undoubtedly take larger.

However, templates are significantly more powerful and obey C++ syntactical rules, where as macros do not.

Template take more compilation time than macros. So, macros are more efficient in term of time.

15. When the inheritance is private, the private methods in base class are in the derived class (in C++).

- (a) inaccessible (b) accessible
- (c) protected (d) public

Ans. (a) : Inaccessible in the derived Class, only public or protected members are inherited

So answer is (a)

16. An assertion is a predicate expressing a condition we wish database to always satisfy. The correct syntax for Assertion is :

- (a) CREATE ASSERTION 'ASSERTION Name' CHECK 'Predicate'
- (b) CREATE ASSERTION 'ASSERTION Name'
- (c) CREATE ASSERTION, CHECK Predicate
- (d) SELECT ASSERTION

Ans. (a) : CREATE ASSERTION 'ASSERTION Name' CHECK 'Predicate' is the proper syntax for applying constraints.

For example : This SQL statement creates an assertion to demand that there's no more than a single president among the employees.

Create assertion AT-MOST-ONE-PRESIDENT as CHECK

(Select count (*) from EMPe where e job = PRESIDENT +4) < 1)

17. Which of the following concurrency protocol ensures both conflict serializability and freedom from deadlock?

- (1) z - phase Locking
- (2) Time stamp - ordering
- (a) Both (a) and (b)
- (b) (1) only
- (c) (2) only
- (d) Neither (1) nor (2)

Ans. (c) : In basic two phase locking there is a chance for deadlock.

Conservative 2 phase is deadlock free

So answer is (2) only

18. Drop Table cannot be used to drop a Table referenced byconstraint.

- (1) Primary key (2) Sub key
- (3) Super key (4) Foreign key
- (a) (1) (b) (1), (2) and (3)
- (c) (4) (d) (1) and (4)

Ans. (c) : If you drop child table first, the foreign key will be dropped as well. If you attempt to drop parent table first you will get an "could not drop object "a" because it is referenced by a foreign key constraint error, Drop table can not be used to drop a table reference by a foreign key constraint. The referencing foreign key constraint or the referencing table must be dropped.

19. Database applications were built directly on top of file system to overcome the following drawbacks of using file-systems :

- (1) Data redundancy and inconsistency
- (2) Difficulty in accessing Data
- (3) Integrity problems

- (a) (1) (b) (1) and (4)
- (c) (1), (2) and (3) (d) (1), (2), (3) and (4)

Ans. (d) : Traditional file system has all mentioned drawbacks. Database applications built by using tables are used to overcome them.

eg : Normalization and integrity constraints reduces redundancy & inconsistency.

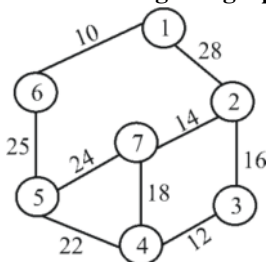
Select * from table provide easy access to data references, joins etc removes data is action primary & foreign key along with check predicates solves integrity problem

20. For a weak entity set to be meaningful, it must be associated with another entity set in combination with some of their attribute values, is called as :

- (a) Neighbor Set (b) Strong Entity Set
- (c) Owner Entity Set (d) Weak Set

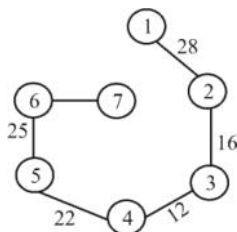
Ans. (c) : Every weak entity must be associated with an identifying entity i.e. the weak entity set is said to be existence dependent on the identifying set. The identifying entity set is said to own the weak entity set that it identifies. It is also called owner entity set.

21. Consider the given graph

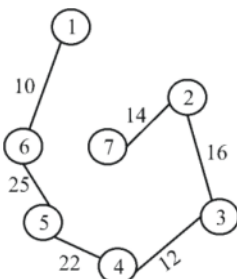


Its Minimum cost spanning tree is

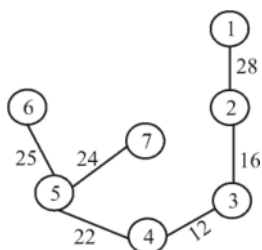
(a)



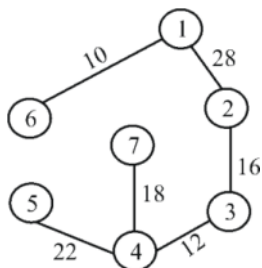
(b)



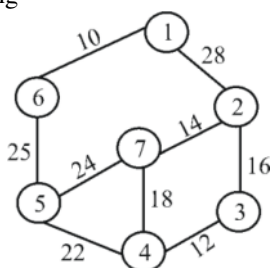
(c)



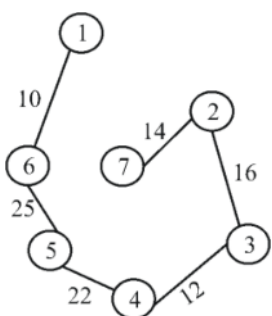
(d)



Ans. (b) : Using



MST :



Option (b) = MST : $10 + 12 + 14 + 16 + 22 + 25 = 99$

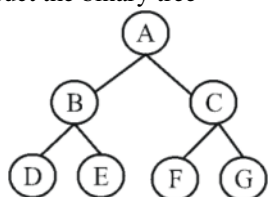
So the minimum is option (b), hence the answer is (b)

22. The in order and preorder Traversal of binary tree are dbefcg and abdecfg respectively. The post-order Traversal is

- (a) dbefacg (b) debfagc
(c) dbefcga (d) debfgca

Ans. (a) : Take the first node in preorder traversal a will be the root.

Now we construct the binary tree



Therefore, postorder is - debfgca

23. Level order Traversal of a rooted Tree can be done by starting from root and performing :

- (a) Breadth first search (b) Depth first search
(c) Root search (d) Deep search

Ans. (a) : Level order traversal is similar to breadth first search whereas preorder traversal is similar to depth first search.

So answer is (a)

24. The average case occurs in the Linear search Algorithm when

- (a) The item to be searched is in some where middle of the array
(b) The item to be searched is not in the array
(c) The item to be searched is in the last of the array
(d) The item to be searched is either in the last or not in the array

Ans. (a) : Linear search runs in at worst linear time and make at most n comparisons, where n is the length of the list. If each element is equally likely to be searched then the linear search has an overage case of $(n/2)$ comparisons, but the average case can be affected if the search probability for each element vary.

25. To determine the efficiency of an algorithm the time factor is measured by:

- (a) Counting micro seconds
(b) Counting number of key operations
(c) Counting number of statements
(d) Counting kilobytes of algorithm

Ans. (b) : The complexity is commonly estimated by counting the number of elementary operations performed by the algorithm, where an elementary operation takes a fixed amount of time to perform. Thus the amount of time taken and the number of elementary operations performed by the algorithm differ by at most a constant factor.

26. Which of the following protocols is an application layer protocol that establish, manages and terminates multimedia sessions?

- (a) Session maintenance protocol
(b) Real – time streaming protocol
(c) Real – time transport control protocol
(d) Session Initiation protocol

Ans. (d) : Session initiation protocol is an application layer control protocol that can establish, modify & terminate multimedia sessions such as internet telephony calls.

27. Match the following port numbers with their uses :

List - I		List - II	
(A)	23	(i)	World wide web
(B)	25	(ii)	Remote login
(C)	80	(iii)	USENET news
(D)	119	(iv)	E-mail

Codes :

	A	B	C	D
(a)	(iv)	(i)	(ii)	(iii)
(b)	(ii)	(i)	(iv)	(iii)
(c)	(ii)	(iv)	(iii)	(i)
(d)	(ii)	(iv)	(i)	(iii)

Ans. (d) :

23 port number is (TELNET) that uses by remote login

25 port number is (SMTP) that uses by E-mail

80 port number is (HTTP) that uses by world wide web

119 port number is used by USENET news

28. Which of the following is not associated with the session layer?

- (a) Dialog control
- (b) Token management
- (c) Semantics of the information transmitted
- (d) Synchronization

Ans. (c) : Semantics of the information transmitted is not associated with the session layer but associated with presentation layer

So answer is (c)

29. What is the size of the 'total length' in IPv4 datagram?

- (a) 4 bits
- (b) 8 bits
- (c) 16 bits
- (d) 32 bits

Ans. (c) : Total length 16 bit field define the entire packet size in IPv4, including header and data, in bytes. The minimum length packet is 20 bytes (20 byte header. + 0 byte data) and maximum is 65, 535 bytes the maximum value of a 16 bit word.

30. Which of the following is/are restriction (s) in classless addressing?

- (a) The number of addresses needs to be power of 2
- (b) The mask needs to be include in the address to define the block
- (c) The starting address must be divisible by the number of addresses in the block
- (d) All of the above

Ans. (d) : In class addressing : (1) the number of addresses needs to be power of 2

(2) The mask needs to be included in the address to define the block

(3) The starting address must be divisible by the number of addresses in block

So answer is (d)

31. Match the following :

List – I		List - II	
(A)	Forward Reference Table	(i)	Assembler directive
(B)	Mnemonic Table	(ii)	Uses array data structure

(C)	Segment Register Table	(iii)	Contains machine OP code
(D)	EQU	(iv)	Uses linked list data structure

Codes :

	A	B	C	D
(a)	(ii)	(iii)	(iv)	(i)
(b)	(iii)	(iv)	(ii)	(i)
(c)	(iv)	(i)	(iii)	(ii)
(d)	(iv)	(iii)	(ii)	(i)

Ans. (d) :

- (a) Forward reference table uses linked list data structure so it satisfies (iv)
 (b) Mnemonic table contains machine op code, so it satisfies (iii)
 (c) Segment register table uses array data structure so it satisfies (ii)
 (d) EQO satisfies (i)

32. The translator which performs macro calls expansion is called :

- (a) Macro processor
 (b) Micro pre-processor
 (c) Macro pre-processor
 (d) Dynamic Linker

Ans. (c) : Translator which performs macro calls expansion is called macro pre-processors

33. If all the production rules have single non-terminal symbol on the left side, the grammar defined is :

- (a) context free grammar
 (b) context sensitive grammar
 (c) unrestricted grammar
 (d) phrase grammar

Ans. (a) : Context free grammar is a set of recursive rewriting rules used to generate patterns of strings. It is consisting of finite set of grammar rules is a quadruple & if all the production rule have single non-terminal symbol on the left side, the grammar defined is context free grammar

i.e. $V \rightarrow (T + V)^+$

34. Which one from the following is false?

- (a) LALR parser is Bottom – up parser
 (b) A parsing algorithm which performs a left to right scanning and a right most deviation is RL (1)
 (c) LR parser is Bottom - Up parser
 (d) In LL (1), the 1 indicates that there is a one - symbol look - ahead.

Ans. (b) : There is nothing RL (1) parsing we have LR (1) parsing algorithm where L stands for left to right scanning of the input string & R signifies that we will get a right most derivation. LALR & LR are bottom up parsers.

So answer (b)

35. Which phase of compiler generates stream of atoms?
- (a) Syntax Analysis (b) Lexical Analysis
(c) Code Generation (d) Code Optimization

Ans. (a) : Syntax analysis is the second phase of a compiler. A syntax analyzer takes the input from a lexical analyzer in the form of token streams. The parser analyzes the source code against the production rules to detect any errors in the code & the output of the phase is stream of atoms.

36. A disk drive has 100 cylinders, numbered 0 to 99. Disk requests come to the disk driver for cylinders 12, 26, 24, 4, 42, 8 and 50 in that order. The driver is currently serving a request at cylinder 24. A seek takes 6 msec per cylinder moved. How much seek time is needed for shortest seek time first (SSTF) algorithm?
- (a) 0.984 sec (b) 0.396 sec
(c) 0.738 sec (d) 0.42 sec

Ans. (d) : Given requests is: 12, 26, 24, 4, 42, 8, 50
Using SSTF (shortest seek time first) algorithm:
Total distance covered
= 2 + 22 + 46 = 70
Total seek time is:
= 70 × 6 msec
= 70 × 6 × 10⁻³ sec = 0.42 sec

37. Let P_i and P_j be two processes, R be the set of variables read from memory, and W be the set of variables written to memory. For the concurrent execution of two processes P_i and P_j which of the following conditions is not true?
- (a) $R(P_i) \cap W(P_j) = \Phi$
(b) $W(P_i) \cap R(P_j) = \Phi$
(c) $R(P_i) \cap R(P_j) = \Phi$
(d) $W(P_i) \cap W(P_j) = \Phi$

Ans. (c) : Reading of two process won't create any problem,
So answer is (c)

38. A LRU page replacement is used with four page frames and eight pages. How many page faults will occur with the reference string 0172327103 if the four frames are initially empty?
- (a) 6 (b) 7
(c) 5 (d) 8

Ans. (b) :									
1	2	3	4	5	hit	hit	hit	6	7
0	0	0	0	3	3	3	3	0	0
	1	1	1	1	1	1	1	1	1
		7	7	7	7	7	7	7	7
			2	2	2	2	2	2	3
Total Page fault = 7									

39. What does the following command do?

grep – vn "abc" x

- (a) It will print all of the lines in the file x that match the search string "abc"
- (b) It will print all of the lines in file x that do not match the search string "abc"
- (c) It will print the total number of lines in the file x that match the string "abc"
- (d) It will print the specific line numbers of the file x in which there is a match for string "abc"

Ans. (b) : Grep (globally search a regular expression & print it) with –V (verbose) option display those line which do not contain the specified string.

So answer is (b)

40. The Unix Kernel maintains two key data structures related to processes, the process table and the user structure. Which of the following information is not the part of user structure?

- (a) File descriptor table
- (b) System call state
- (c) Scheduling parameters
- (d) Kernel stack

Ans. (c) : The unix kernal maintains two key data structures related to processes, the process table and the user structure. In user structure there is no place for scheduling parameter process table stores the information of all running process. The process table contains process ID's, memory usage of the process, what are file descriptor used in the process. So, option (c) is correct.

41. Match the following :

List - I		List - II	
(A)	Size-oriented metrics	(i)	Uses number of external interfaces as one of the measurement parameter
(B)	Function-oriented metrics	(ii)	Originally designed to be applied to business information system
(C)	Extended function point metrics	(iii)	Derived by normalizing quality and/or productivity measures by considering the size of the software
(D)	Function point	(iv)	uses algorithm characteristics as one of the measurement parameter

Codes :

	A	B	C	D
(a)	(iii)	(iv)	(i)	(ii)
(b)	(ii)	(i)	(iv)	(iii)
(c)	(iv)	(ii)	(iii)	(i)
(d)	(iii)	(i)	(iv)	(ii)

Ans. (d) :

- (a) Size oriented metrics is directly implied with (iii)
- (b) Function oriented metrics is number of user inputs, number of user outputs, number of files, number of external interfaces. So it satisfies (i)
- (c) Extended function point metrics is a superset of the function point measure that can be applied to system. It uses algorithm characteristics as one of the measurement parameter.
- (d) Function point is a "unit of measurement" to express the amount of business functionality an information provides to a user.

42. In which testing strategy requirements established during requirements analysis are validated against developed software?

- (a) Validation testing
- (b) Integration testing
- (c) Regression testing
- (d) System testing

Ans. (a) : Validation is "Are we building the right product? According to the capability maturity model
Software validation : The process of evaluating software during or at the end of the development process to determine whether it satisfies specified requirements.

43. Which process model is also called as classic life cycle model?

- (a) Waterfall model
- (b) RAD model
- (c) Prototyping model
- (d) Incremental model

Ans. (a) : The waterfall model was first process model to be introduce.

- It is also referred as linear sequential life cycle model
- It is very simple to understand and use. In this each phase must be complicated fully before the next phase can begin.
- In this model the testing stands only after the development is complete
- In this model phases do not overlap

44. Cohesion is an extension of :

- (a) Abstraction concept
- (b) Refinement concept
- (c) Information hiding concept
- (d) Modularity

Ans. (c) : A cohesion module performs a single task within a software procedure, requiring little interaction with little procedures in other parts of software.

Type of cohesion :

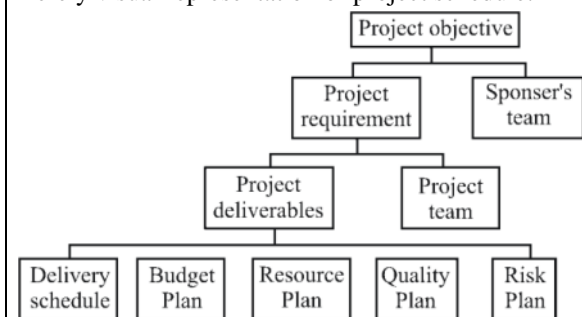
- A module that performs tasks that are related logically is logically cohesion

- When a module contains tasks that are related by the fact that all must be executed with the same span of time, the exhibits temporal cohesion
- At the low end of the spectrum, a module that performs a set of tasks that relate to each other loosely called coincidentally cohesive.

45. Which one from the following is highly associated activity of project planning?

- Keep track of the project progress
- Compare actual and planned progress and costs
- Identify the activities, milestones and deliverables produced by a project
- Both (b) and (c)

Ans. (c) : Project planning is commonly perceived as creating 'Gantt chart', which is incorrect. Gantt chart is merely visual representation of project schedule.



46. In the case of parallelization, Amdahl's law states that if P is the proportion of a program that can be made parallel and (1-P) is the proportion that cannot be parallelized, then the maximum speed-up that can be achieved by using N processors is :

- $\frac{1}{(1-P) + N \cdot P}$
- $\frac{1}{(N-1)P + P}$
- $\frac{1}{(1-P) + \frac{P}{N}}$
- $\frac{1}{P + \frac{(1-P)}{N}}$

Ans. (c) : Amdahl's law states that if P is the proportion of a program that can be made parallel & (1 - P) is the proportion that cannot be parallelized, then the maximum speed up that can be achieved by using N processors is.

$$\frac{1}{(1-P) + \frac{P}{N}}$$

47. Which of the following statement is incorrect for parallel virtual machine (PVM)?

- The PVM communication model asynchronous blocking send, asynchronous blocking receive, and non-blocking receive function.
- Message buffers are allocated dynamically
- The PVM communication model assumes that any task can send a message to any other

PVM task and that there is no limit to the size or number of such messages

- (d) In PVM model, the message order is not preserved.

Ans. (d) : Fault tolerance through dynamic addition & deletion of hosts/processes.

A widely used parallel message passing library defines

- How takes communicative or synchronize with each other

48. Which of the following algorithms sort n integers, having the range 0 to $(n^2 - 1)$, in ascending order in $O(n)$ time?

- (a) Selection sort (b) Bubble sort
(c) Radix sort (d) Insertion sort

Ans. (c) : Radix sort take $O(n)$ time, so answer is (c)

49. Which of the following statements is FALSE about weak entity set?

- (a) Weak entities can be deleted automatically when their strong entity is deleted
(b) Weak entity set avoids the data duplication and consequent possible inconsistencies caused by duplicating the key of the strong entity
(c) A weak entity set has no primary keys unless attributes of the strong entity set on which it depends are included
(d) Topples in a weak entity set are not partitioned according to their relationship with topples in a strong entity set

Ans. (d) : We have weak entities for several reasons :

- We want to avoid the data duplication and consequent entity. Possible inconsistencies caused by duplicating the key of strong.
- Weak entities be stored physically with their strong entities
- Weak entities can be detected automatically when their strong entity is detected.

So (d) is incorrect statement, Hence (d) is the answer.

50. Which of the following is not valid with reference to message passing interface (MPI)?

- (a) MPI can run on any hardware platform
(b) The programming model is a distributed memory model
(c) All parallelism is implicit
(d) MPI - Comm. - size returns the total number of MPI processes in specified communication

Ans. (c) : Message passing interface (MPI) is a standardized and portal message passing system.

- It can run on any hardware platform. It is portable, efficient, flexible.
- It has become a de facto standard for communication among memory system.
- MPI - Comm. - Size returns the total number of MPI processes in specified communication.

So answer is (c)